Enhancing Critical Thinking: Exploring Human-AI Synergy in Student Cognitive Development

Abstract

Artificial intelligence (AI) has emerged as a transformative tool, integrated across various sectors. In education, AI has generated significant excitement among students for its potential to enhance learning experiences. However, concerns about overreliance on AI temper this enthusiasm, as it may undermine the development of critical thinking skills. Numerous studies have highlighted the risks associated with students' excessive use of generative AI (GAI) in academic tasks, noting its potential to diminish cognitive abilities. However, the optimal use of AI to enhance students' critical thinking skills remains under-researched. Therefore, this study seeks to answer the question: How does generative AI influence students' critical thinking, self-efficacy, and decision-making? This study aims to explore the synergic relationship between human intelligence and artificial intelligence in augmenting essential thinking skills among students and building upon their existing cognitive resources through self-efficacy, learning motivation, and decision-making. Specifically, it explores the cause-and-effect connections among GAI, self-efficacy, decision-making, learning motivation, and critical thinking skills. A quantitative methodology used an online questionnaire to collect responses from 165 undergraduate, master's, and doctoral students. Statistical analyses, including bootstrapping techniques, were conducted to examine direct and indirect effects. The results revealed that GAI has a significant positive influence on self-efficacy, learning motivation, decision-making, and critical thinking skills. In turn, self-efficacy, learning motivation, and decision-making significantly impact critical thinking skills. The mediating results indicated that GAI can indirectly boost students' critical thinking by enhancing self-efficacy, learning motivation, and decision-making. This suggests that AI capabilities can transform the cognitive learning process.